

Souvenirs

Amaru ari kugura souvenirs mu iduka ryo mu mahanga. Hari amoko N ya souvenirs. Umubare wa buri bwoko bwa souvenir buri mu iduka ntago ushira.

Ikiguzi cya buri bwoko ntago gihinduka. Bishatse kuvuga, souvenir y' ubwoko i ($0 \leq i < N$) igura ibiceri $P[i]$, where $P[i]$ is a positive integer.

Amaru azi neza ko amoko ya souvenir yaranje mu buryo bugabanuka mu giciro, kandi ko nta bwoko bugura kimwe n' ubundi. Bishatse kuvuga, $P[0] > P[1] > \dots > P[N-1] > 0$.

Ikindi kandi, yashoboye kumenya igiciro cy' ubwoko bwa $P[0]$, gusa ntago yashoboye kumenya ibiciro by' ayandi moko asigaye.

Kugira ngo agure souvenirs, Amaru azakora transactions nyinshi n' umucuruzi.

Buri transaction igizwe na steps zikurikira:

1. Amaru aha umucuruzi ibiceri runaka.
2. Umucuruzi arabifata akabitondekanya ku meza ahantu Amaru adashobora kubona.
3. Umucuruzi areba buri bwoko bwa souvenir ahereye kuri $0, 1, \dots, N-1$, imwe kuri imwe.
Buri bwoko abureba inshuro **imwe**.
 - Mu gihe ari kureba ubwoko bwa souvenir i , iyo afite nibura ibiceri $P[i]$ cyangwa hejuru yabyo
 - umucuruzi akura ibiceri bingana nka $P[i]$ muri rwa rutonde, yaragiza
 - agashyira souvenir y' ubwoko i ku meza.
4. Nyuma y' ibyo, Umucuruzi afata ibiceri byose bisigaye na souvenirs yasize ku meza akabiha Amaru.

Ugomba kumenya ko nta bicere cyangwa souvenirs byari biri ku meza y' umucuruzi mbere y' uko umucuruzi atangira iyi transaction.

Akazi kawo ni ugutegeka Amaru gukora transactions runaka, ku buryo:

- kuri buri transaction agura **nibura souvenir imwe**, kandi
- Muri rusange agura souvenirs **zingana nka** i z' ubwoko i , kuri buri i aho $0 \leq i < N$. Menya ko bishatse kuvuga ko Amaru atangomba kugura souvenir y' ubwoko 0.

Amaru ntago atengetswe gukoresha umubare muto ushoboka wa transactions.

Implementation Details

Ugomba gushyira mu bikorwa ibi bikurikikira.

```
void buy_souvenirs(int N, long long P0)
```

- N : umubare w' amoko ya souvenirs.
- $P0$: igiciro cya $P[0]$.
- Iyi function ihamagarwa inshuro imwe kuri buri test case.

Muri function ya `buy_souvenirs` ashobora guhamagara iyi function ikurikira kugira ngo akore transaction:

```
std::pair<std::vector<int>, long long> transaction(long long M)
```

- M : Ibiceri Amaru yahaye umucuruzi.
- Iyi function ireturninga a pair. element yambere ya pair ni array L , irimo amoko ya souvenirs yaguzwe (in increasing order). Element ya kabiri ni integer R , umubare w' ibiceri byagaruriwe Amaru nyuma ya transaction.
- Ni ngombwa ko $P[0] > M \geq P[N - 1]$. Iyi condition $P[0] > M$ ituma Amaru atangura souvenir y' ubwoko 0, kandi $M \geq P[N - 1]$ ituma Amaru agura nibura souvenir imwe. Mu gihe izo conditions zitubahirijwe solution yawe ibona `Output isn't correct: Invalid argument`. Ibuka ko uretse $P[0]$, ibiciro bya $P[N - 1]$ ntago biba byatanzwe muri input.
- Transaction ihamagarwa inshuro zitarenze 5000 kuri buri test case.

Uburyo grader ikosora **ntago buhindurwa na inputs**. Bishatse kuvuga ko urutonde rw' ibiciro P ruba ruzwi mbere yuko `buy_souvenirs` ihamagarwa.

Constraints

- $2 \leq N \leq 100$
- $1 \leq P[i] \leq 10^{15}$ kuri buri i aho $0 \leq i < N$.
- $P[i] > P[i + 1]$ kuri buri i aho $0 \leq i < N - 1$.

Subtasks

Subtask	Score	Additional Constraints
1	4	$N = 2$
2	3	$P[i] = N - i$ kuri buri i aho $0 \leq i < N$.
3	14	$P[i] \leq P[i + 1] + 2$ kuri buri i aho $0 \leq i < N$.
4	18	$N = 3$
5	28	$P[i + 1] + P[i + 2] \leq P[i]$ kuri buri i aho $0 \leq i < N - 2$. $P[i] \leq 2 \cdot P[i + 1]$ kuri buri i aho $0 \leq i < N - 1$.
6	33	Nta zindi constraints.

Example

Nk' urugero

```
buy_souvenirs(3, 4)
```

Hari amoko $N = 3$ ya souvenirs kandi $P[0] = 4$. Nureba neza urasanga hari uburyo butatu bw' ibiciro P : $[4, 3, 2]$, $[4, 3, 1]$, and $[4, 2, 1]$.

Tekereza ko `buy_souvenirs` ihamagaye `transaction(2)`. tekereza ko `transaction` ireturninze $([2], 1)$, bishatse kuvuga ko amaru yaguze souvenir imwe y' ubwoko 2 kandi ko umucuruzi yamugaruriye igiceri 1. Nureba neza urasanga ibi bituma twanzura ko $P = [4, 3, 1]$, kubera ko:

- Iyo $P = [4, 3, 2]$, `transaction(2)` yari kuba yagaruye $([2], 0)$.
- Iyo $P = [4, 2, 1]$, `transaction(2)` yari kuba yagaruye $([1], 0)$.

Noneho `buy_souvenirs` yahamagara `transaction(3)`, `transaction` ikagarura $([1], 0)$, bishatse kuvuga ko Amaru aguze souvenir imwe y' ubwoko 1 kandi ko umucuruzi yamugaruriye ibicirero 0. Kugeza ubu, muri rusange, Yaguze souvenir imwe y' ubwoko 1 n' indi y' ubwoko 2.

Finally, `buy_souvenirs` yahamagara `transaction(1)`, ikagarura $([2], 0)$, bishatse kuvuga ko Amaru aguze souvenir y' ubwoko 2. Menya ko hano dushobora kuba twakoresheje na `transaction(2)`. Iki gihe, muri rusange Amaru afite souvenir imwe y' ubwoko 1 n' ebyiri z' ubwoko 2, nkuko byasabwe.

Sample Grader

Input format:

N

$P[0] \quad P[1] \quad \dots \quad P[N-1]$

Output format:

$Q[0] \quad Q[1] \quad \dots \quad Q[N-1]$

Hano $Q[i]$ ni umubare wa souvenirs z' ubwoko i bwaguzwe muri rusange kuri buri i such that $0 \leq i < N$.